

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. [Organic Chemistry Tutor: Ohm's Law](#)
 - Conventional current is positive to negative
 - Resistors drop voltage
 - KVL: makes the sum of all voltages around any closed loop in a circuit equal zero.
 - Parallel each branch gets the same volts and different amps
 - Inversely proportional for Resistance and Current
 - KCL: Current that enters a junction is the same as Current that leaves it
2. [Khan Academy, Current, Resistors, Ohm's Law:](#)
 - $V = IR$ or $I = V/R$
 - Voltage is electric potential
 - Water pressure
 - Current is charge per unit of time
 - How much water flows
 - Resistance is impedance of charge
 - Its like a narrowing in a pipe
 - Conventional current positive to negative
 - Real electrons go from negative to positive
3. [Khan Academy, Compound Circuits:](#)
 - Across a resistor, the charges decrease their electric potential energy
 - This turns into thermal energy in the resistor
 - Power = change in electrical potential energy / time
 - $U = \text{charge} * \text{Voltage}$
 - Watts = Joules per second
 - Power can also be $P = I^2 * R$
 - $P = \Delta v \text{ squared} / R$
4. [EEVBlog, Voltage Dividers:](#)
 - Output always less than input

- Norton Equivalent
 - Single current source in parallel with a single resistor
 - Thevenin Equivalent
 - Single voltage source in series with a resistor
5. [EEVBlog, Kirchhoff's Laws:](#)

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWare Power Electronics Course:
 - a. [Lecture 5: Intro to DC/DC, Part 1](#)
2. [Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic](#)
3. [Power Electronics by Daniel W. Hart](#)

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!